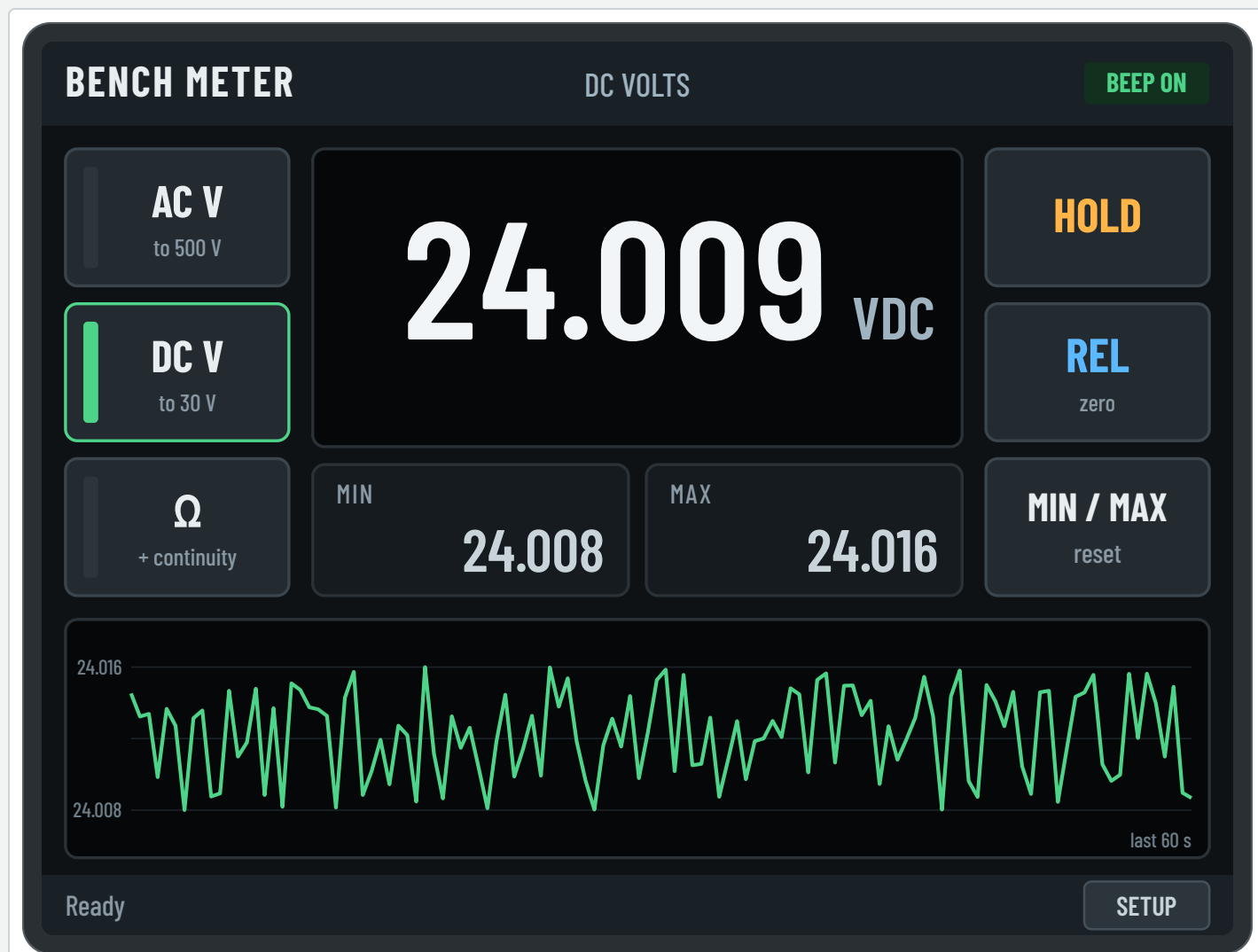


Bench Meter Screens

Two screens for the EA7-T8C: the main meter and a setup screen. The mockups below work like the real thing, so tap the buttons to see how each mode, HOLD, REL and MIN/MAX behave. The tables give every object's position, size, tag and settings to build it in the C-more Programming Software.

Screen 1 · Meter



Try it: tap a mode, HOLD, REL or MIN/MAX. The mockup uses sample readings, not live data.

☐ simulate over-range

Screen 1 · Objects

Positions are X, Y of the top-left corner and W × H, in pixels. Colors are suggestions; the reading only has to stay high-contrast.

OBJECT (C-MORE TYPE)	X, Y · W × H	TAG / ADDRESS	SETTINGS
BACKGROUND & HEADER			
Screen background	—	—	Screen color  #0E1215
Rectangle (header bar)	0, 0 · 800 × 56	—	 #1A2126
Static Text "BENCH METER"	16, 10	—	28 pt bold, white
Lookup Text (mode name)	300, 12 · 200 × 34	DS1 MODE	1 = "AC VOLTS", 2 = "DC VOLTS", 3 = "OHMS", centred, 22 pt
Indicator Text "HOLD"	560, 14 · 68 × 28	C20 HOLD_ON	On: amber text on dark amber. Off: invisible (same as background).
Indicator Text "REL"	636, 14 · 56 × 28	C21 REL_ON	On: blue on dark blue. Off: invisible.
Indicator Text "BEEP ON / OFF"	700, 14 · 84 × 28	C5 BEEP_EN	On: "BEEP ON" green. Off: "BEEP OFF" grey.
MODE BUTTONS (LEFT)			
Pushbutton "AC V"	16, 72 · 150 × 92	C6 HMI_ACV	Momentary On. Lamp tag C51: lamp on = accent border / bright face.
Pushbutton "DC V"	16, 176 · 150 × 92	C7 HMI_DCV	Momentary On. Lamp tag C52.
Pushbutton "Ω"	16, 280 · 150 × 92	C8 HMI_OHM	Momentary On. Lamp tag C53.
READING (CENTRE)			
Rectangle (reading well)	182, 72 · 436 × 200	—	 #050708, 2 px border #2A3238
Numeric Display (AC)	196, 96 · 340 × 120	DF30 DISPLAY	Float, 4 whole digits · 1 decimal · right-aligned · ~110 pt · visible when C51 on
Numeric Display (DC)	196, 96 · 340 × 120	DF30 DISPLAY	Same spot · 2 whole digits · 3 decimals · visible when C52 on
Numeric Display (Ω)	196, 96 · 340 × 120	DF30 DISPLAY	Same spot · 6 whole digits · 1 decimal · visible when C53 on

OBJECT (C-MORE TYPE)	X, Y · W × H	TAG / ADDRESS	SETTINGS
Lookup Text (units)	540, 160 · 70 × 50	DS1 MODE	1 = "VAC", 2 = "VDC", 3 = "Ω", 40 pt, right-aligned
Indicator Text "OL"	196, 96 · 408 × 120	C44 SHOW_OL	On: red "OL", 120 pt, with a solid background that covers the number . Off: invisible. Place it on top of the three displays.
MIN / MAX			
Rectangle + Static Text "MIN"	182, 284 · 212 × 88	—	#151B1F, label 18 pt grey
Numeric Display (min)	250, 316 · 132 × 48	DF32 MIN_VAL	2 decimals, 40 pt, right-aligned
Rectangle + Static Text "MAX"	406, 284 · 212 × 88	—	as above
Numeric Display (max)	474, 316 · 132 × 48	DF33 MAX_VAL	2 decimals, 40 pt, right-aligned
FUNCTION BUTTONS (RIGHT)			
Pushbutton "HOLD"	634, 72 · 150 × 92	C1 HMI_HOLD	Momentary On. Lamp tag C20 (lit while holding).
Pushbutton "REL"	634, 176 · 150 × 92	C2 HMI_REL	Momentary On. Lamp tag C21.
Pushbutton "MIN / MAX reset"	634, 280 · 150 × 92	C3 HMI_MMRESET	Momentary On
TREND & STATUS			
Trend Graph	16, 388 · 768 × 160	DF30 DISPLAY	Sample every 0.5 s · 120 samples (60 s) · auto-scale off. Set the range per mode, or use 0–500 as a compromise. Line #4ED488.
Rectangle (status bar)	0, 560 · 800 × 40	—	#1A2126
Indicator Text (excitation)	16, 566 · 420 × 28	C13 EXCITE_FLT	On: "Ω excitation lost — check F4 / PTC2" in red
Indicator Text (VACT loop)	16, 566 · 420 × 28	C14 LOOP_FLT	On: "VACT loop open — check TB6 wiring" in red. Visible only in AC mode (C51).

OBJECT (C-MORE TYPE)	X, Y · W × H	TAG / ADDRESS	SETTINGS
Screen Selector "SETUP"	700, 564 · 84 × 32	—	Go to screen 2. Optionally password-protect it.

Screen 2 · Setup

SETUP

DC calibration

Factor = reference volts ÷ CH2 volts

3.0021

tap to edit
range 2.9 – 3.1

Continuity beeper

Beeps below 30 Ω, stops above 35 Ω

ON

Raw inputs (volts at the module)

CH1 · VACT loop	4.842	expect 1.00 at 0 VAC
CH2 · DC divider	7.998	× factor = DC volts
CH3 · volts across Rx	3.334	≈ 3.33 with 1 kΩ
CH4 · excitation	10.002	should be ≈ 10.0

◀ METER

CLICK Port 2 · 38400 8-0-1 · node 1

OBJECT (C-MORE TYPE)	X, Y · W × H	TAG / ADDRESS	SETTINGS
Numeric Entry (DC factor)	32, 148 · 200 × 72	DF20 DCV_CAL	Float, 1 whole digit · 4 decimals · min 2.9, max 3.1 , so a typo can't wreck the scaling
Toggle Pushbutton	424, 148 · 160 × 72	C5 BEEP_EN	Toggle · on "ON" green, off "OFF" grey
Numeric Display ×4	250, 324 · 110 × 28 (step 40)	DF1 · DF2 · DF3 · DF4	3 decimals, right-aligned, 20 pt

OBJECT (C-MORE TYPE)	X, Y · W × H	TAG / ADDRESS	SETTINGS
Static Text ×8	labels at x 32 and x 376	–	Channel names and expected values, as shown
Screen Selector "◀ METER"	16, 494 · 200 × 60	–	Go to screen 1

Project setup

New project

1. C-more Programming Software → New Project → panel **EA7-T8C**.
2. **Panel Manager:** protocol *AutomationDirect CLICK* on the serial port. 38400 baud, odd parity, 8 data bits, 1 stop bit, PLC address 1, RS-232.
3. Screen 1 = Meter, and make it the **start-up screen**. Screen 2 = Setup.
4. Set the screen saver to dim rather than off, or turn it off, so a bench reading never disappears mid-test.
5. Transfer the project over USB, then connect the EA-2CBL to CLICK **Port 2**. Port 1 stays free for the programming cable.

Tag database

- **Floating PT 32:** DF1–DF4, DF20, DF30, DF32, DF33
- **Signed Int 16:** DS1
- **Discrete:** C1–C3, C5–C8, C13, C14, C20, C21, C44, C51–C53
- Use the ladder nicknames as the tag names (MODE, DISPLAY, HOLD_ON...), so the panel and PLC programs read the same.
- If a Numeric Display shows odd huge numbers, the tag was probably made as an integer. Change it to Floating PT 32.